FALCON · origin-to-destination scenarios

10 scenarios · run of 2026-11-03

AT A GLANCE

What this run shows

- **Origin to destination.** The best route between two ports, with up to 3 routes per move, each through different waters. 10 moves between 12 ports in 8 countries.
- **Realistic sea routes.** Routes follow a published shipping-lane network through real straits and canals, not straight lines (Eurostat MARNET: 9,686 lane segments, 22 named straits and canals). This run's routes pass through 7 named straits.
- **Area denial and what-if scenarios.** 12 denial areas in 9 scenarios (7 circles of 40 to 150 nm radius and 5 shapes drawn on the map). Every shipping lane that enters an area is closed, the routes are found again, and the cost is measured in distance and in sailing time at 18 knots. 1 scenario shows FALCON recognising that no route remains, and saying why.

Mode tested: Sea. Air moves between airfields, along great-circle routes, are planned.

How it was tested

- All 10 scenarios passed all 110 automated checks.
- 57 checks compare each answer with what its scenario expects: which straits the best route uses or avoids, whether the closure touches the normal route, and how much longer the route gets.
- 53 checks hold every route to rules it must always keep: it starts and ends at its ports, never enters a closed area, is never shorter than the straight-line distance, and its drawn line matches its reported length.

What it found

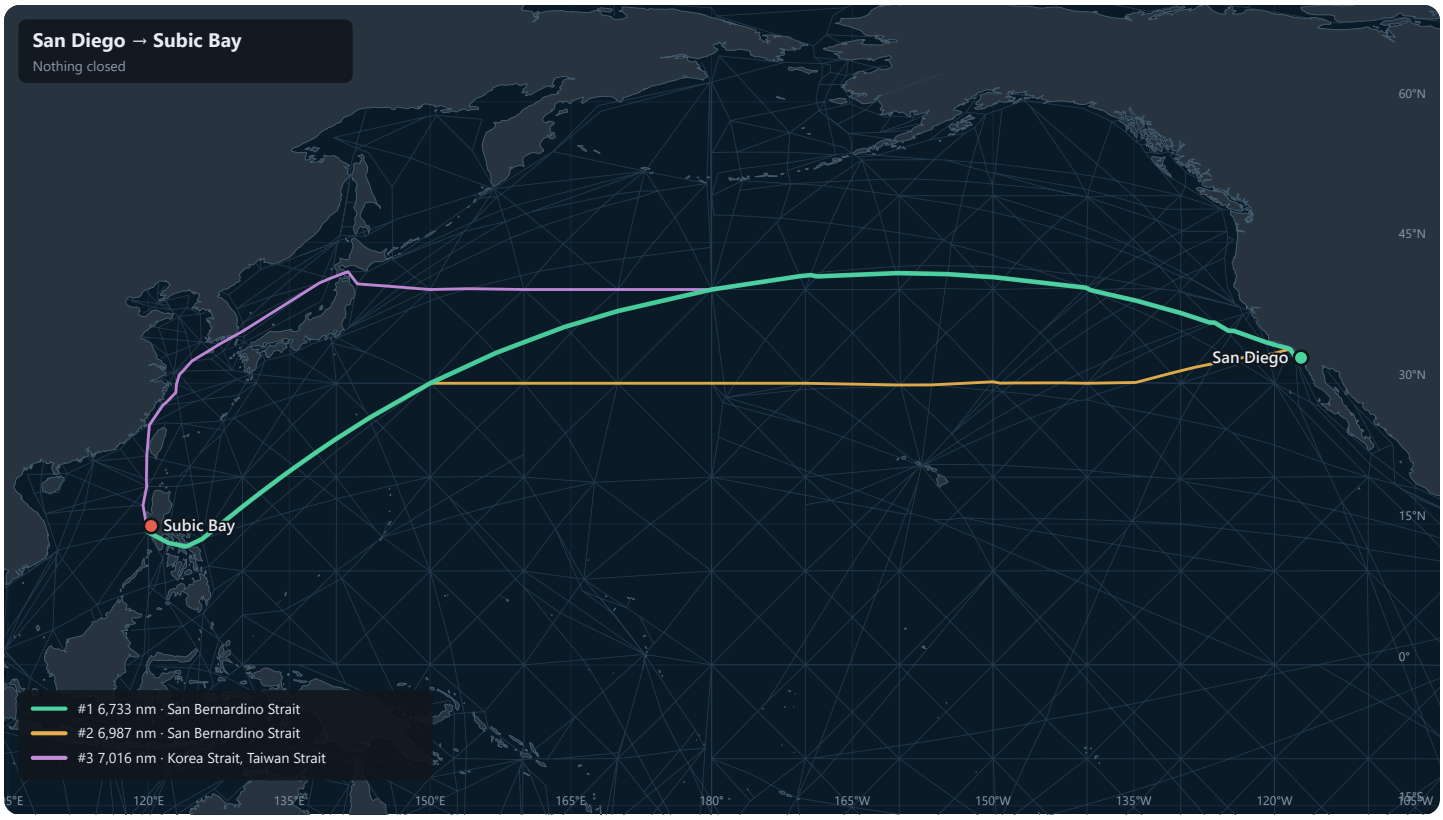
- Largest relative cost: Manila to Puerto Princesa, Mindoro Strait closed (65% longer, 407 to 672 nm).
- Longest delay: Diego Garcia to Yokosuka, Malacca, Sunda and Lombok Straits closed (about 2.8 days more at 18 knots).
- Smallest effect: San Diego to Subic Bay, San Bernardino Strait closed (0.5% longer).
- 1 closure leaves no route at all: Honolulu to Kao-Hsiung, Taiwan inside a closed area.

WHAT EACH CLOSURE COSTS

SCENARIO	NORMAL (NM)	WITH CLOSURE (NM)	EXTRA	EXTRA TIME
Manila to Puerto Princesa, Mindoro Strait closed	407	672	+265 nm (65%)	15 hours
Singapore to Manila, central South China Sea closed	1,342	2,150	+808 nm (60%)	1.9 days
Diego Garcia to Yokosuka, Malacca, Sunda and Lombok Straits closed	5,359	6,567	+1,208 nm (23%)	2.8 days
Colombo to Yokosuka, Strait of Malacca closed	4,595	5,183	+588 nm (13%)	1.4 days
Guam to Manila, Luzon and San Bernardino Straits closed	1,699	1,864	+165 nm (9.7%)	9 hours
Sasebo to Guam, waters from Taiwan to southern Kyushu closed	1,545	1,659	+114 nm (7.4%)	6 hours
Yokosuka to Singapore, Luzon Strait closed	2,951	2,977	+26 nm (0.9%)	1 hour
San Diego to Subic Bay, San Bernardino Strait closed	6,733	6,768	+35 nm (0.5%)	2 hours
Honolulu to Kao-Hsiung, Taiwan inside a closed area	4,711	No route: Destination port Kao-Hsiung is inside a closed area.		

San Diego to Subic Bay, open seas

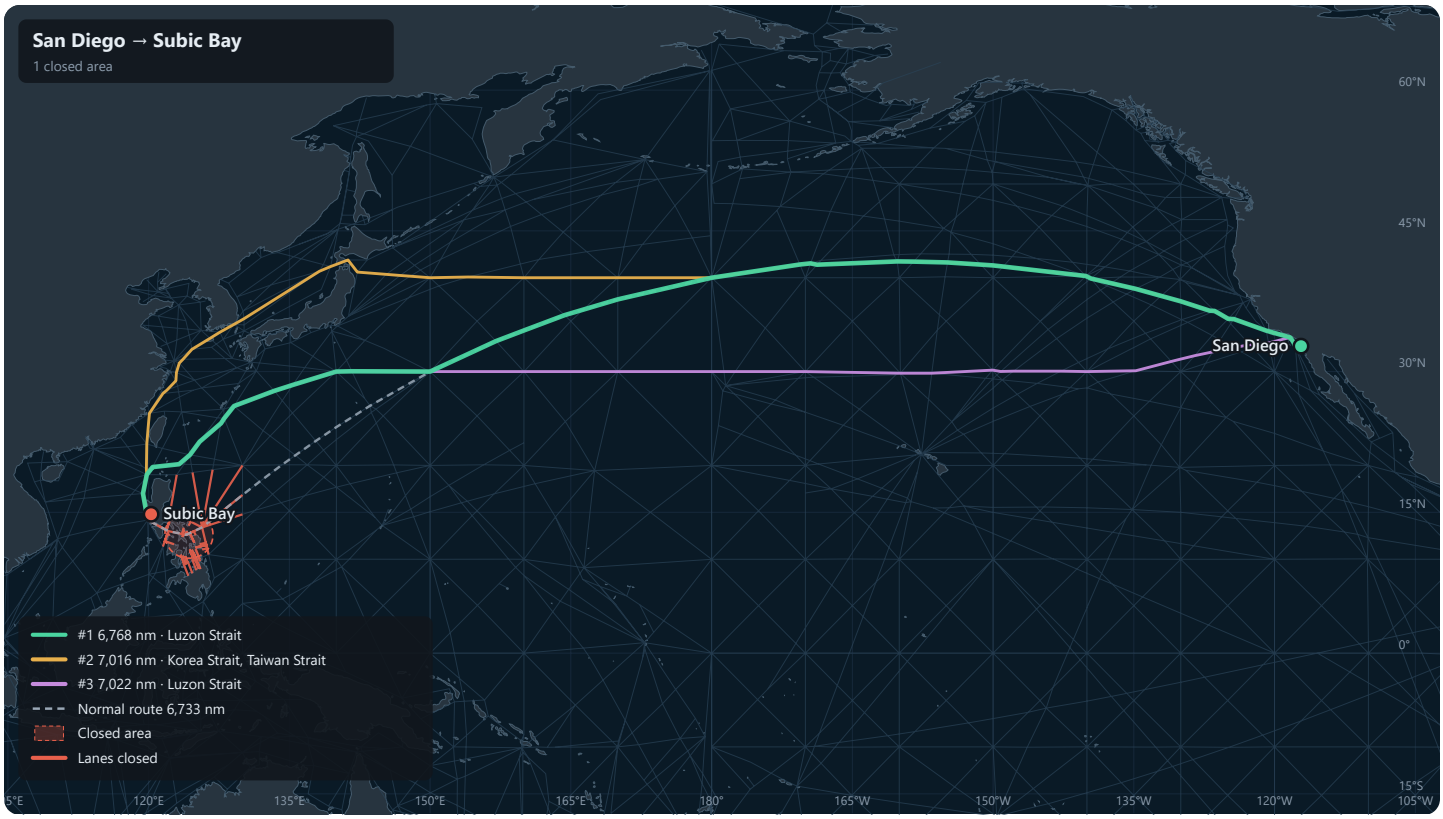
With nothing closed, the best route is 6,733 nm via San Bernardino Strait, about 15.6 days at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Best route	6,733	15.6	San Bernardino Strait
Alternative 2	6,987	16.2	San Bernardino Strait
Alternative 3	7,016	16.2	Korea Strait, Taiwan Strait

San Diego to Subic Bay, San Bernardino Strait closed

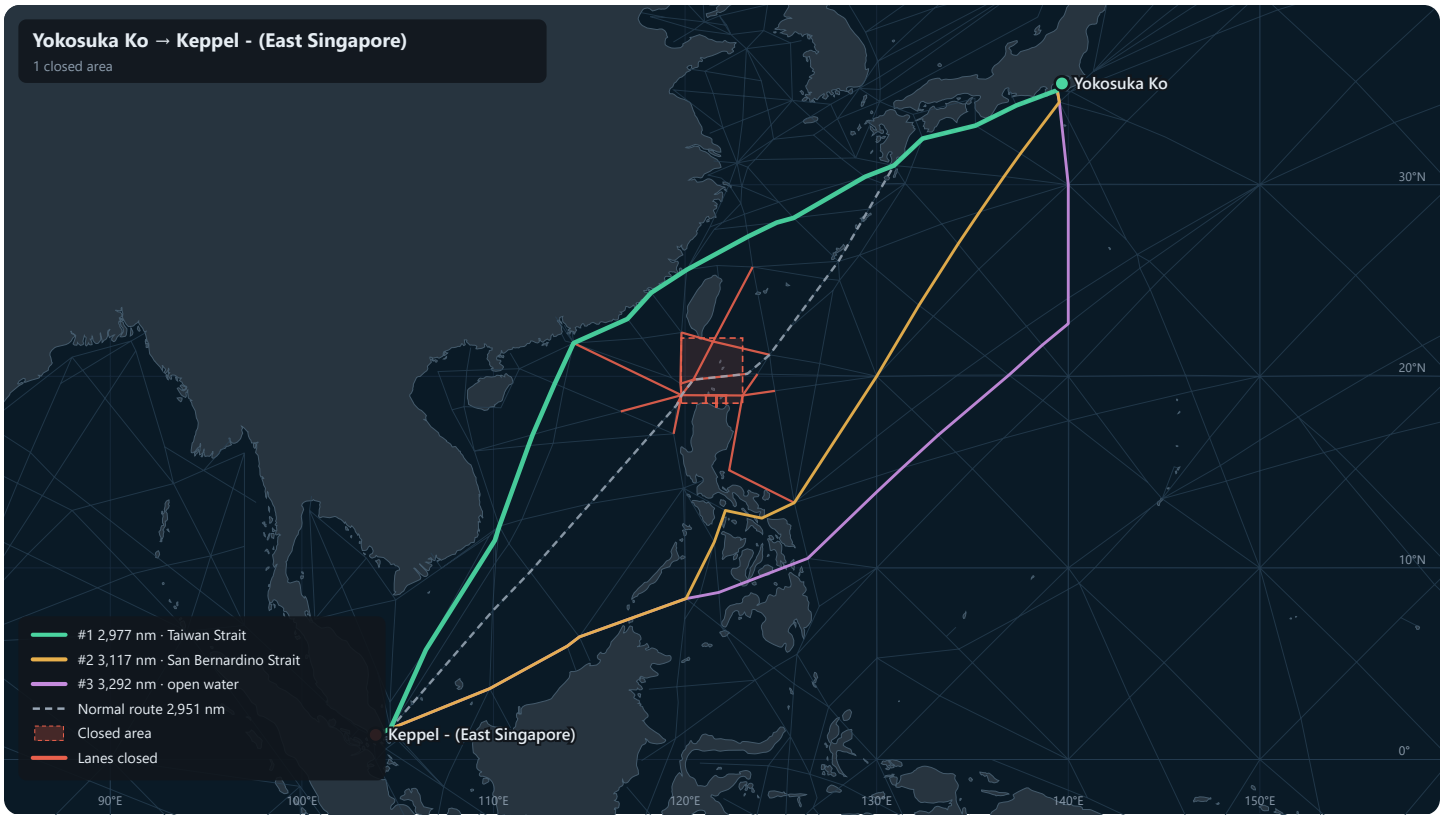
With the closed area in place, the best route is 6,768 nm via Luzon Strait, instead of 6,733 nm via San Bernardino Strait. That is 35 nm (0.5%) longer: about 2 hours more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	6,733	15.6	San Bernardino Strait
Best route with the closure	6,768	15.7	Luzon Strait
Alternative 2	7,016	16.2	Korea Strait, Taiwan Strait
Alternative 3	7,022	16.3	Luzon Strait

Yokosuka to Singapore, Luzon Strait closed

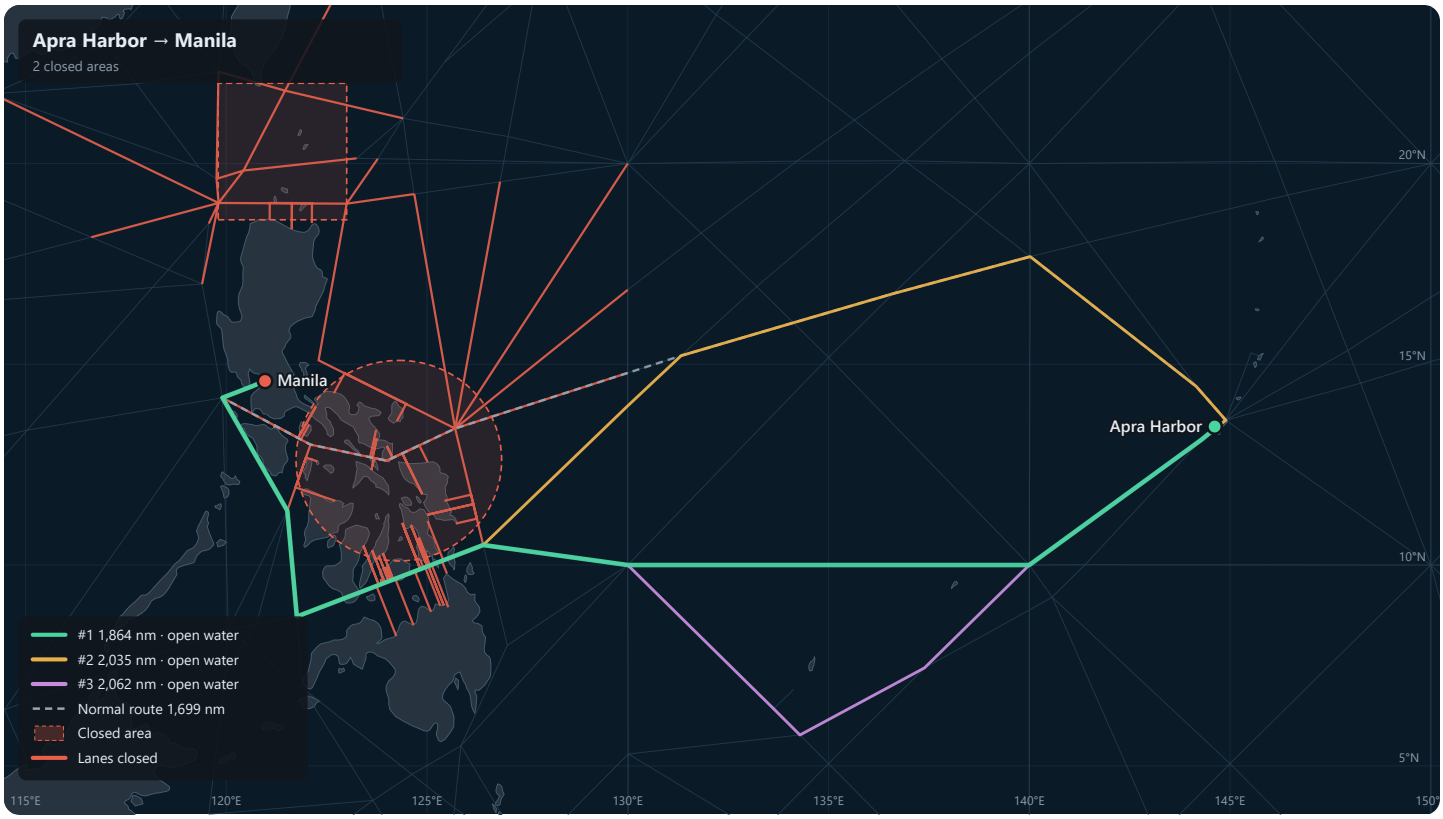
With the closed area in place, the best route is 2,977 nm via Taiwan Strait, instead of 2,951 nm via Luzon Strait. That is 26 nm (0.9%) longer: about 1 hour more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	2,951	6.8	Luzon Strait
Best route with the closure	2,977	6.9	Taiwan Strait
Alternative 2	3,117	7.2	San Bernardino Strait
Alternative 3	3,292	7.6	open water

Guam to Manila, Luzon and San Bernardino Straits closed

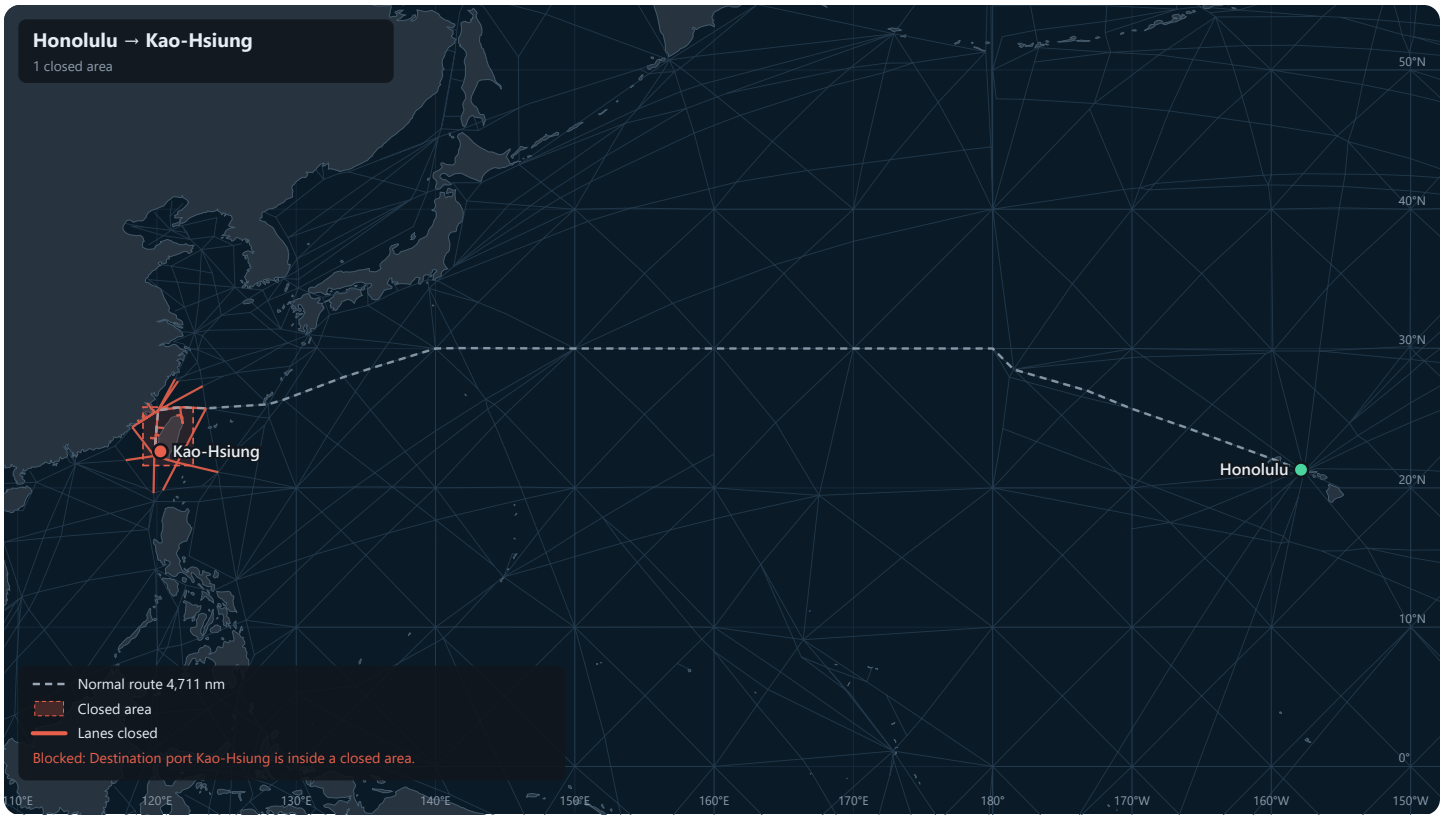
With the 2 closed areas in place, the best route is 1,864 nm through open water, instead of 1,699 nm via San Bernardino Strait. That is 165 nm (9.7%) longer: about 9 hours more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	1,699	3.9	San Bernardino Strait
Best route with the closure	1,864	4.3	open water
Alternative 2	2,035	4.7	open water
Alternative 3	2,062	4.8	open water

Honolulu to Kao-Hsiung, Taiwan inside a closed area

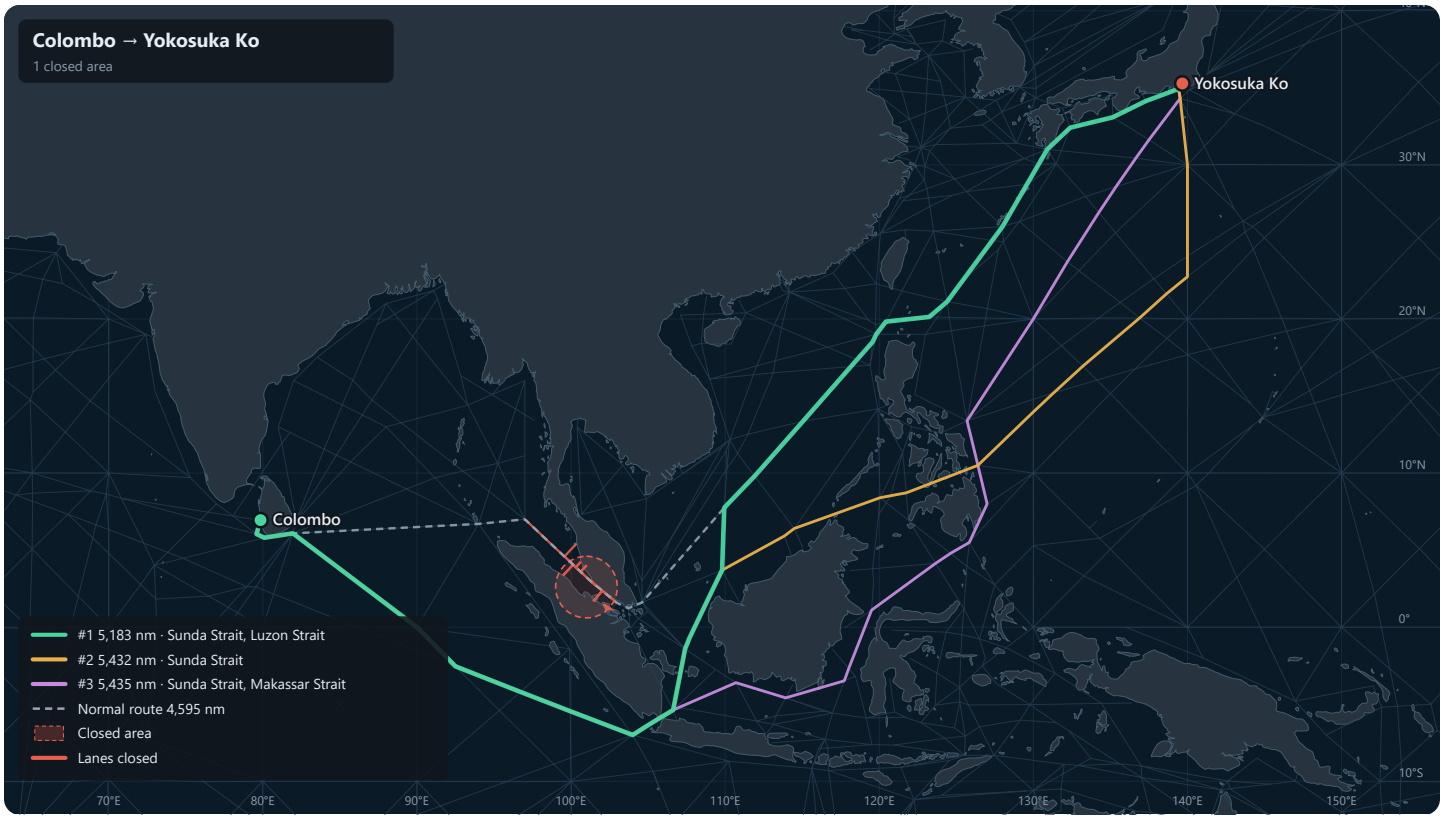
No route is possible. Destination port Kao-Hsiung is inside a closed area. With nothing closed, the route is 4,711 nm via Taiwan Strait.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	4,711	10.9	Taiwan Strait

Colombo to Yokosuka, Strait of Malacca closed

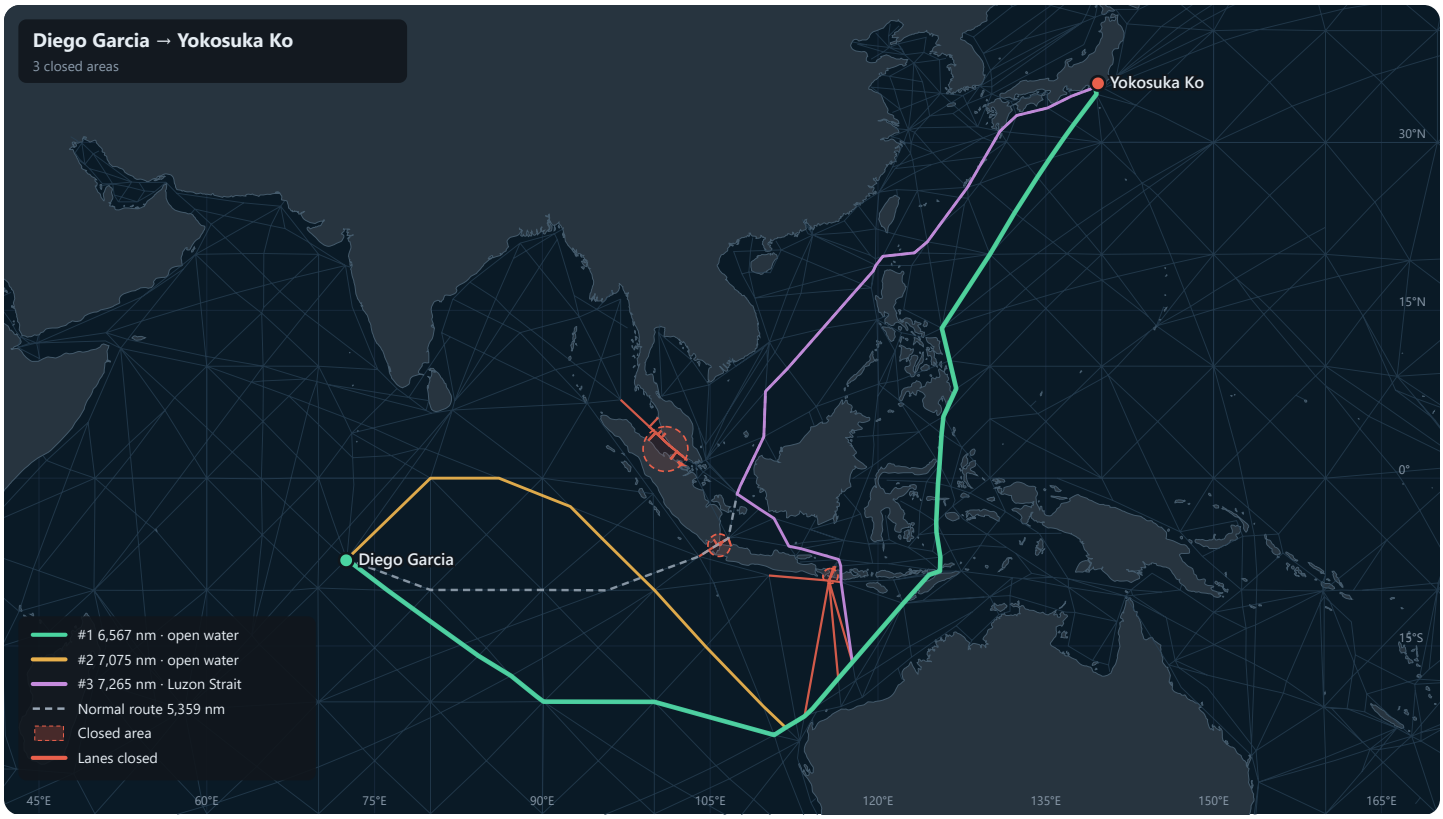
With the closed area in place, the best route is 5,183 nm via Sunda Strait, Luzon Strait, instead of 4,595 nm via Strait of Malacca, Luzon Strait. That is 588 nm (13%) longer: about 1.4 days more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	4,595	10.6	Strait of Malacca, Luzon Strait
Best route with the closure	5,183	12.0	Sunda Strait, Luzon Strait
Alternative 2	5,432	12.6	Sunda Strait
Alternative 3	5,435	12.6	Sunda Strait, Makassar Strait

Diego Garcia to Yokosuka, Malacca, Sunda and Lombok Straits closed

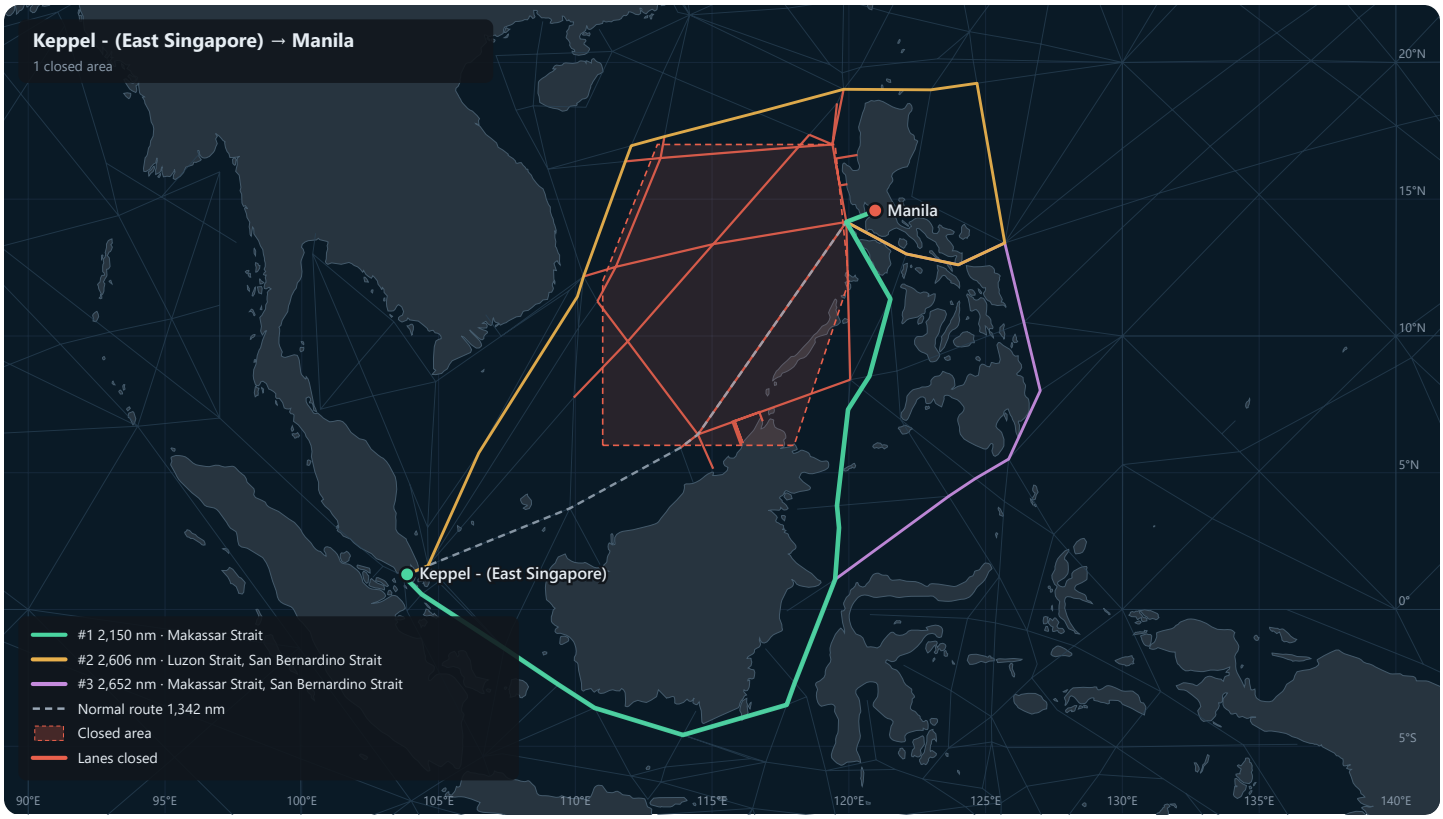
With the 3 closed areas in place, the best route is 6,567 nm through open water, instead of 5,359 nm via Sunda Strait, Luzon Strait. That is 1,208 nm (23%) longer: about 2.8 days more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	5,359	12.4	Sunda Strait, Luzon Strait
Best route with the closure	6,567	15.2	open water
Alternative 2	7,075	16.4	open water
Alternative 3	7,265	16.8	Luzon Strait

Singapore to Manila, central South China Sea closed

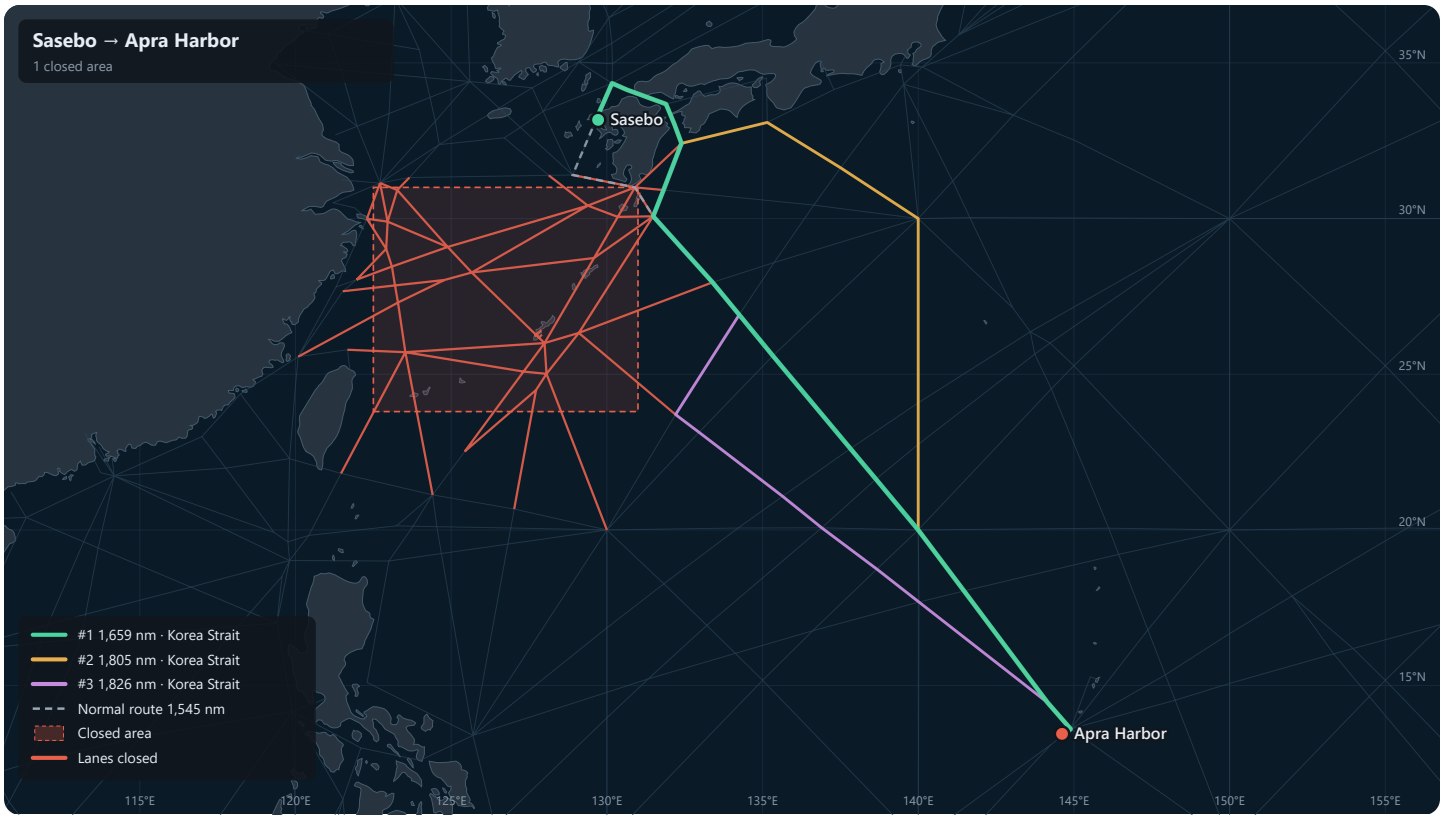
With the closed area in place, the best route is 2,150 nm via Makassar Strait, instead of 1,342 nm through open water. That is 808 nm (60%) longer: about 1.9 days more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	1,342	3.1	open water
Best route with the closure	2,150	5.0	Makassar Strait
Alternative 2	2,606	6.0	Luzon Strait, San Bernardino Strait
Alternative 3	2,652	6.1	Makassar Strait, San Bernardino Strait

Sasebo to Guam, waters from Taiwan to southern Kyushu closed

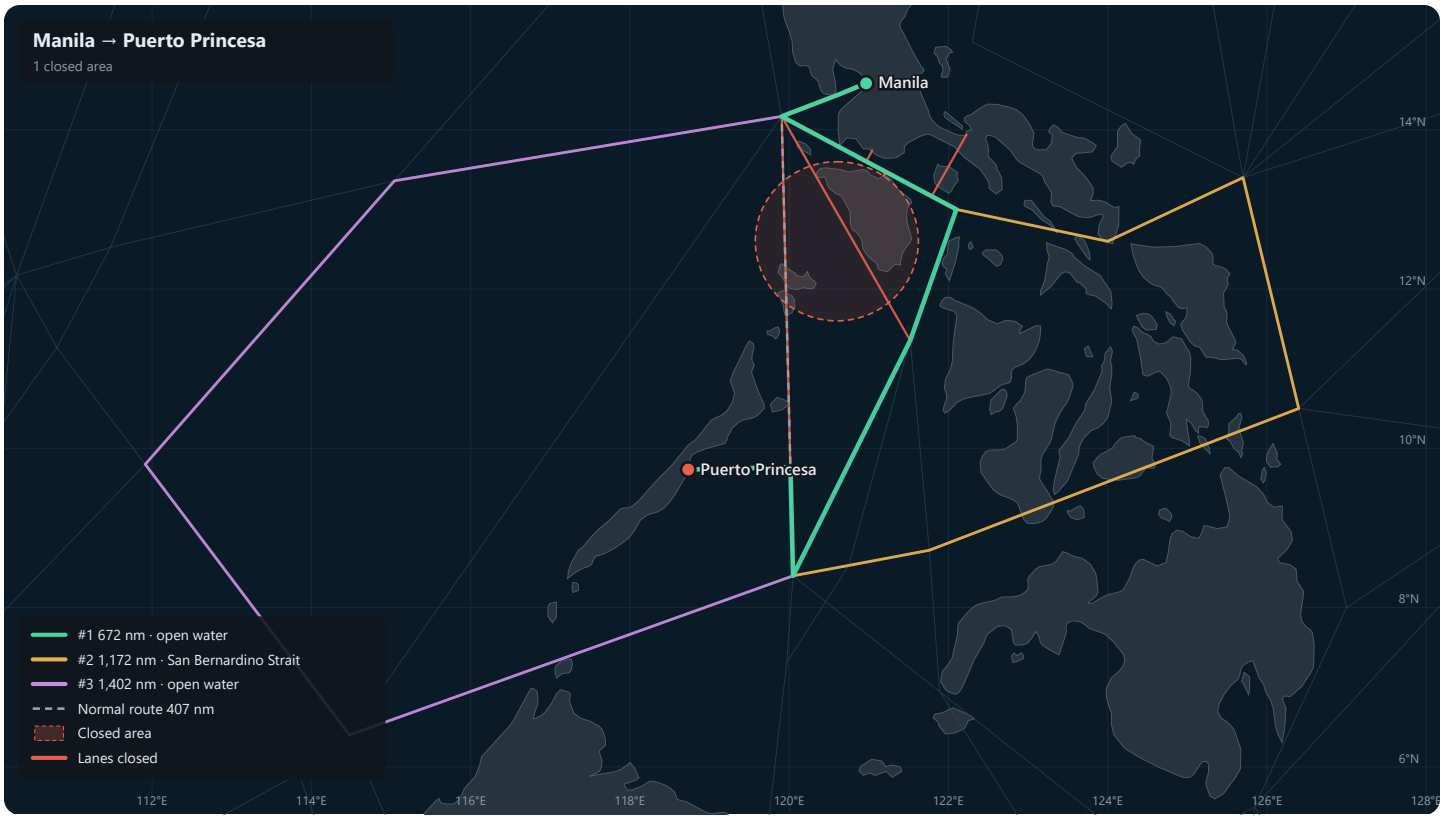
With the closed area in place, the best route still goes via Korea Strait, but is 1,659 nm instead of 1,545 nm. That is 114 nm (7.4%) longer: about 6 hours more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	1,545	3.6	Korea Strait
Best route with the closure	1,659	3.8	Korea Strait
Alternative 2	1,805	4.2	Korea Strait
Alternative 3	1,826	4.2	Korea Strait

Manila to Puerto Princesa, Mindoro Strait closed

With the closed area in place, the best route still goes through open water, but is 672 nm instead of 407 nm. That is 265 nm (65%) longer: about 15 hours more at 18 knots.



OPTION	DISTANCE (NM)	DAYS AT 18 KT	VIA
Normal route, nothing closed	407	0.9	open water
Best route with the closure	672	1.6	open water
Alternative 2	1,172	2.7	San Bernardino Strait
Alternative 3	1,402	3.2	open water

ASSUMPTIONS AND LIMITS

- These results compare what-if routes. They do not predict or detect events.
- Distances are the shortest path along a published shipping-lane network (Eurostat MARNET, about 100 km detail), including the short leg from each port to its nearest lane. Real voyages also depend on weather, traffic and routing rules that are not modelled.
- Time is distance at a constant 18 knots, for a rough sense of scale only.
- A denial area closes every shipping lane that enters it. How or for how long an area is closed is not modelled.
- 1,877 of 3,806 ports reach their nearest lane by an approximate straight leg of 10 to 50 nm, and 500 by one of more than 50 nm, so distances to and from those ports are rougher.
- Port depth, port size and vessel limits are not yet applied: any port can start or end a move.
- FALCON's distances have not yet been checked against published port-to-port distance tables.

DATA SOURCES

DATA	SOURCE
Ports	NGA World Port Index (Pub. 150)
Shipping lanes	Eurostat searoute, MARNET network
Coastline (maps only)	Natural Earth, 1:50m land

From the scenario run of 2026-11-03 10:56:11 UTC-0500, FALCON code ebe1c62. All 10 scenarios passed 110 automated checks; the full QA report is report.html in the same folder.